

I was unable to get the project working on maven

Links to the commits of the separate feature branches:

Feature 1:

<https://github.com/Cjimen47/CSE4642026-CJimen47/compare/feature1...main>

Feature 2:

<https://github.com/Cjimen47/CSE4642026-CJimen47/commits/feature2/>

Feature 3:

<https://github.com/Cjimen47/CSE4642026-CJimen47/commits/feature3/>

Feature 4:

<https://github.com/Cjimen47/CSE4642026-CJimen47/commits/feature4/>

Project Part 2

How To Interact with the code

System will ask you to input the file path of your graph, input 'CSE464PJ/userGraph.dot'

This is the expected output

```
Welcome to the Graph Manager
Please enter the name of the file containing the graph
CSE464PJ/userGraph.dot
Here is the graph you inputed:
Number of Nodes: 8
Label of Nodes: [a, b, c, d, e, f, g, h]
Number of Edges: 9
Nodes and Edge Direction of Edges: {
a->b;
b->c;
c->d;
d->a;
a->e;
e->f;
e->g;
f->h;
g->h;
}
```

System will ask you to input the source and destination node to find a bfs path for, input 'a,h'

This is the expected output

```
To find a breadth first search path, please enter a source and destination node (separate by comma):  
a,h  
Here is the path found:  
a->b->e->c->f->g->d->h
```

System will ask you to input the source and destination node to find a dfs path for, input 'a,h'

This is the expected output

```
Here is the path found:  
a->b->c->d->e->f->h
```

System will ask you to input the node you wish to remove, input 'h'

This is the expected output

```
Please enter the node you wish to remove:  
h  
Node h Successfully removed.  
Here is the resulting graph:  
Number of Nodes: 7  
Label of Nodes: [a, b, c, d, e, f, g]  
Number of Edges: 7  
Nodes and Edge Direction of Edges: {  
a->b;  
b->c;  
c->d;  
d->a;  
a->e;  
e->f;  
e->g;  
}
```

System will ask you to input the nodes you wish to remove, input 'd,f,g'

This is the expected output

```
Please enter the nodes you wish to remove (separate by comma):
```

```
d,f,g
```

```
Node d successfully removed.
```

```
Node f successfully removed.
```

```
Node g successfully removed.
```

```
Here is the resulting graph:
```

```
Number of Nodes: 4
```

```
Label of Nodes: [a, b, c, e]
```

```
Number of Edges: 3
```

```
Nodes and Edge Direction of Edges: {
```

```
a->b;
```

```
b->c;
```

```
a->e;
```

```
}
```

System will ask you to input the source and destination nodes of the edge you wish to remove, input 'b,c'

This is the expected output

```
Please enter the source and destination of the edge you wish to remove (separate by comma):
```

```
b,c
```

```
Edge b -> c has been successfully removed
```

```
Here is the resulting graph:
```

```
Number of Nodes: 4
```

```
Label of Nodes: [a, b, c, e]
```

```
Number of Edges: 2
```

```
Nodes and Edge Direction of Edges: {
```

```
a->b;
```

```
a->e;
```

```
}
```

Github Commits:

Remove Node, Remove Nodes, and Remove Edges Commits:

<https://github.com/Cjimen47/CSE4642026-CJimen47/commits/rmvAPI>

Breadth First Search:

<https://github.com/Cjimen47/CSE4642026-CJimen47/commits/bfs>

Depth First Search:

<https://github.com/Cjimen47/CSE4642026-CJimen47/commits/dfs>

Main:

<https://github.com/Cjimen47/CSE4642026-CJimen47/commits/main>